



Dr. Akkem Yaganteeswarudu

Senior Data scientist at Deloitte

Ph.D., NIT Silchar

Postdoc fellow, SR University , Warangal

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SUMMARY:

Strategic AI Leader with over 15 years of experience in Generative AI, Machine Learning, Agentic AI, Deep Learning, and NLP, specialized in bridging the gap between advanced research and enterprise-scale deployment. Ph.D. in Computer Science with a proven track record of managing multi-disciplinary teams (up to 20 to 30 members) to automate core business processes by 40% at Deloitte. Expert in Explainable AI (XAI), Multi-Agent Systems with a portfolio including 1,370+ career citations and granted international patents. Dedicated to building ethical, transparent, and high-performance AI solutions.

CORE COMPETENCIES

- **Leadership:** AI Strategy & Roadmap Development, Multi-Disciplinary Team Management (15+ members), Cross-Functional Collaboration, Budget & Resource Forecasting.
- **Generative AI & Agentic Systems:** RAG Frameworks, Multi-Agentic Orchestration (LangGraph), LLM Fine-tuning (GPT-4, Gemini), Prompt Engineering Governance.
- **Advanced Analytics:** Explainable AI (SHAP, LIME), Predictive Modeling, Anomaly Detection.
- **Operational Excellence:** Azure AI Studio, Scalable Architecture (FastAPI, AKS), Agile/GitFlow, Performance Metrics.
- **Programming:** Python, Pandas, NumPy, Scikit Learn

EDUCATION:

Course	University/College	Branch	%	Supervisor
Postdoc fellow, 2026	SR University	CSE (AI & ML)		Dr. Muthu Kumar T, Dr. Saroj Kumar Biswas
Ph.D., 2025	NIT, Silchar	CSE	9.25 CGPA	Dr. Saroj Kumar Biswas, Dr. Aruna Varanasi
Master's Degree (M. Tech)	J.N.T.U - SJCTET	CSE	78	Dr. Chandra Sekhar
Bachelor's Degree (B. Tech)	J.N.T.U - SJCTET	CSE	74	Dr. Chandra Sekhar
Intermediate	Srikrishna Jr College	MPC	94	NA
SSC	Govt Boys high school	ALL	84	NA

CERTIFICATIONS:

- **Certified AWS Machine Learning Specialty Professional**
- **Google Cloud Certified – Professional Machine Learning Engineer**
- **Google L400 Highest Badge for GenAI** (Expertise in **RAG Frameworks** and **Generative AI Solution Deployment**)
- **Microsoft Certified Solution Developer / Associate**
- **Hacker Rank (Python Basics) Certified**

Publications

International Journals:

- **Akkem, Y.**, Biswas, S. K., & Aruna, V. (2025). Deciphering the black box: interactive crop recommendation system using Explainable AI with visualisation dashboards. *Journal of Experimental & Theoretical Artificial Intelligence*, 1–41. <https://doi.org/10.1080/0952813X.2025.2595024>
- **Yaganteeswarudu Akkem**, Saroj Kumar Biswas, Varanasi Aruna 2025. AI-Driven Smart Farming Portal: Overcoming Language Barriers and Enhancing Agricultural Productivity through Machine Learning. *Engineering Research Express* in press <https://doi.org/10.1088/2631-8695/ade19d>
- **A Yaganteeswarudu**, Saroj Kumar Biswas, Varanasi Aruna, Deeksha Tripathi, Enhancing Transparency in Smart Farming: Local Explanations for Crop Recommendations Using LIME, *Procedia Computer Science*, Volume 258,2025, Pages 1993-2005, ISSN 1877-0509, <https://doi.org/10.1016/j.procs.2025.04.450>.
- **Yaganteeswarudu Akkem**, Saroj Kr. Biswas, "Analysis of An Intellectual Mechanism of a Novel Crop Recommendation System using Improved Heuristic Algorithm-based Attention and Cascaded Deep Learning Network" in *IEEE Transactions on Artificial Intelligence*, vol. 1, no. 01, pp. 1-14, Aug. 5555, doi: 10.1109/TAI.2024.3508654.
- **Akkem Yaganteeswarudu**, Biswas, S.K. & Varanasi, A. Streamlit-based enhancing crop recommendation systems with advanced explainable artificial intelligence for smart farming. *Neural Computing & Applications* 36, 20011–20025 (2024). <https://doi.org/10.1007/s00521-024-10208-z>.
- **Yaganteeswarudu Akkem**, Saroj Kumar Biswas, Aruna Varanasi, A comprehensive review of synthetic data generation in smart farming by using variational autoencoder and generative adversarial network, *Engineering Applications of Artificial Intelligence*, Volume 131,2024,107881, ISSN 0952-1976, <https://doi.org/10.1016/j.engappai.2024.107881>.
- **Yaganteeswarudu Akkem**, Saroj Kumar Biswas, Aruna Varanasi, Smart farming using artificial intelligence: A review, *Engineering Applications of Artificial Intelligence*, Volume 120,2023,105899, ISSN 0952-1976, <https://doi.org/10.1016/j.engappai.2023.105899>.
- **Yaganteeswarudu Akkem**, Saroj Kumar Biswas, Aruna Varanasi, "Role of Explainable AI in Crop Recommendation Technique of Smart Farming", *International Journal of Intelligent Systems and Applications (IJISA)*, Vol.17, No.1, pp.31-52, 2025. DOI:10.5815/ijisa.2025.01.03
- **Akkem Yaganteeswarudu**, Kumar BS, Varanasi A. (2023) Streamlit Application for Advanced Ensemble Learning Methods in Crop Recommendation Systems – A Review and Implementation. *Indian Journal of Science and Technology*. 16(48): 4688-4702. <https://doi.org/10.17485/IJST/v16i48.2850>.

International Conferences:

- **Akkem, Yaganteeswarudu**, Biswas, S.K., Varanasi, A. (2023). Smart Farming Monitoring Using ML and MLOps. In: Hassanien, A.E., Castillo, O., Anand, S., Jaiswal, A. (eds) *International Conference on Innovative Computing and Communications. ICICC 2023*. Lecture Notes in Networks and Systems, vol 703. Springer, Singapore. https://doi.org/10.1007/978-981-99-3315-0_51.

- **Akkem Yaganteeswarudu**, "Multi Disease Prediction Model by using Machine Learning and Flask API," 2020 5th International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India, 2020, pp. 1242-1246, <https://doi.org/10.1109/ICCES48766.2020.9137896>.
- **Yaganteeswarudu, A.**, Dasari, P. (2021). Diabetes Analysis and Risk Calculation – Auto Rebuild Model by Using Flask API. In: Chen, J.IZ., Tavares, J.M.R.S., Shakya, S., Ilyasu, A.M. (eds) Image Processing and Capsule Networks. ICIPCN 2020. Advances in Intelligent Systems and Computing, vol 1200. Springer, Cham. https://doi.org/10.1007/978-3-030-51859-2_27
- **Yaganteeswarudu, A.**, Varanasi, A., Mohanty, S. (2021). Security in Software Applications by Using Data Science Approaches. In: Shakya, S., Balas, V.E., Haoxiang, W., Baig, Z. (eds) Proceedings of International Conference on Sustainable Expert Systems. Lecture Notes in Networks and Systems, vol 176. Springer, Singapore. https://doi.org/10.1007/978-981-33-4355-9_27
- **A. Yaganteeswarudu** and Y. VishnuVardhan, "Software application to prevent suicides of farmers with asp.net MVC," 2017 7th International Conference on Cloud Computing, Data Science & Engineering - Confluence, Noida, India, 2017, pp. 543-546, doi: 10.1109/CONFLUENCE.2017.7943210.
- **A. Yaganteeswarudu**, "The speaking compiler-A compiler with audio for immediate error correction," 2016 3rd International Conference on Computing for Sustainable Global Development (INDIACom), New Delhi, India, 2016, pp. 980-983.
- **A. Yaganteeswarudu** and B. V. Devi, "The Multi Language Audio Compiler with Video Help," 2018 3rd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT), Bangalore, India, 2018, pp. 204-208, doi: 10.1109/RTEICT42901.2018.9012266.

Patents and Patent applications:

➤ **South African Patent**

Title: AN ARTIFICIAL INTELLIGENCE BASED EXPLAINABLE CROP RECOMMENDATION SYSTEM

Status: **Granted**

Countries where it is Filed: **South Africa**

Publication number: **2024/01704**

Field of invention: The present disclosure relates to an artificial intelligence based explainable crop recommendation system, specifically to an explainable crop recommendation system that utilizes an explainable artificial intelligent approach (XAI) to enable an interpretable crop recommendation system.

Inventors: **Yaganteeswarudu Akkem**, Saroj Kumar Biswas, Aruna Varanasi

➤ **Indian Design Patent**

Title: Artificial intelligence-based processing device for providing Smart Farming Crop Recommendations

Application number: **445181-001**

Status: **Granted**

Countries where it is Filed: **India**

Inventors: **Yaganteeswarudu Akkem**, Saroj Kumar Biswas, Aruna Varanasi

Field of invention: Designed a smart farming device to provide accurate crop recommendations

➤ **South African Patent**

Title: SYSTEM AND METHOD FOR PARAMETER-EFFICIENT AND DYNAMICALLY WEIGHTED ENSEMBLE-BASED GENE MUTATION CLASSIFICATION

Status: **Granted**

Countries where it is Filed: **South Africa**

Publication number: **2026/01258**

Inventors: **Yaganteeswarudu Akkem**, Muthu Kumar T, Saroj Kumar Biswas

Thesis Title: Design of Explainable Crop Recommendation System for Smart Farming using Machine Learning

Status: Ph.D. Awarded

Supervisors: Dr. Saroj Kumar Biswas (NIT Silchar) | Dr. Aruna Varanasi (SNIST, Hyderabad)

This research developed a novel, high-performance, and interpretable AI system for real-time crop recommendation, focusing on addressing data scarcity, model opacity, and computational efficiency challenges in predictive agriculture.

➤ **Key Technical Contributions**

- **Novel Machine Learning and Deep Learning Architecture:** Developed an **Ensemble Based Stacking** and **Attention-based Cascaded Deep Learning Network** integrated with stacked ensemble learning. This innovative framework significantly **enhanced model accuracy** while simultaneously **reducing computational time complexity**, a critical factor for real-time deployment.
- **Explainable AI (XAI) for Transparency:** Implemented a hybrid feature selection and interpretation methodology combining **SHapley Additive exPlanations (SHAP)** and **Local Interpretable Model-agnostic Explanations (LIME)**. This provided transparent, variable-level insights into model decisions, improving user trust and enabling **model interpretability** alongside high predictive performance.
- **Synthetic Data Generation (GenAI Concept):** Utilized **Generative Adversarial Networks (GANs)** to create diverse and representative synthetic datasets, successfully mitigating the critical challenge of **labelled data scarcity** in the agricultural domain and substantially augmenting model robustness.
- Designed a comprehensive system architecture for streamlined operation, featuring:
 - Integration with **IoT sensors** for dynamic, real-time context-aware recommendations.
 - An **AI module accelerated by FPGA processors** for executing rapid crop predictions.
 - Deployment and results presentation via a production-ready **Streamlit application**.

➤ **Quantified Results & Novelty**

- **Performance:** Achieved **considerable improvements in precision and accuracy** of crop predictions compared to traditional baseline models through the application of the novel AACNet and ensemble learning methods.
- **Efficiency:** The AACNet framework effectively **reduced time complexity**, enabling quicker data processing crucial for timely agricultural decision-making.
- **Novelty:** Pioneering the application of **GANs for synthetic data generation** and the **SHAP/LIME hybrid for feature selection** within the crop recommendation sector.

Technologies Used: Deep Learning (AACNet, GANs), Explainable AI (SHAP, LIME), Ensemble Learning, IoT Integration, Streamlit.

Work experience in detail:

Deloitte (October 2021 – present)

Managing Data science team and Senior consultant Data scientist

Project Healthcare UMR usecase

Client: Major Healthcare Provider

Project1:

Project:	Healthcare UMR Classification & Alert Generation
Technologies:	GPT-4.1, Dataiku, Snowflake, Python, LLM

Impact & Achievement Focus	
UMR Classification:	Designed and deployed LLM-based pipelines (GPT-4.1) using written prompts to classify Unsolicited Medical Requests (UMRs) into General, Sub, Strategic, and Emerging Themes.
Summarization & Insights:	Utilized LLM summarization to provide concise insights on UMR trends across all classification themes.
Alert Generation & Analysis:	Generated Actionable Alerts with Definitions and Comparative Summaries to identify and track volume change and distribution change in medical requests.
Data & MLOps:	Leveraged Snowflake for data access and Dataiku for pipeline orchestration, ensuring a scalable and robust solution delivery.

Project: Multi Agentic Business Process Analyzer (MAS-BPA)

Technologies:	Python, FastAPI, LangGraph, React, Azure Kubernetes Service (AKS), Azure PostgreSQL, LLMs (GPT-4, Gemini)
Multi-Agentic Solution:	Developed a multi-agentic AI solution that accelerates and automates business process analysis, documentation, and re-imagination.
LangGraph Orchestration:	Engineered a suite of LangGraph agents—including Process Configuration, BPMN Generator, and Documentation—orchestrated with a human-in-the-loop controller.
Automated Artifacts:	Deployed specialized agents to auto-generate critical artifacts such as BPMN 2.0 diagrams, process documentation, and Business Requirement Documents (BRDs).
Performance Diagnostics:	Integrated a Process Health Check module that maps KPI data to activities to identify root causes of underperformance and suggest corrective actions.
Transformation Roadmap:	Leveraged AI to propose future-state designs, automatically generating implementation-ready "To-Be" packages (YAML, BPMN, and documentation).
Scalable Architecture:	Built a robust backend on Azure Kubernetes Service using FastAPI and Azure PostgreSQL, ensuring model flexibility across multiple LLM providers.

Project GenAI BMR usecases

Project: GenAI BMR Use Cases
Technologies: Python, GPT Models (3/4), Lang Chain, Azure AI Studio
Impact & Achievement Focus
GenAI Solution Deployment: Led a 10-member team in the full-stack development and deployment of 3 RAG-based Generative AI solutions for Business Case Modelling and Quantitative Data Synthesis.
Efficiency & Cost Savings: The deployed solutions successfully automated 40% of routine BMR use cases, resulting in an estimated annual saving of 1,200 man-hours across client operations.
MLOps Implementation: Standardized the GenAI solution deployment using the Azure AI Studio MLOps pipeline , reducing the average model-to-production time from 3 weeks to 2 days .
Team Leadership: Directed data science team efforts, implementing rigorous code review processes that ensured a 95% code quality standard and GitFlow compliance across all GenAI repositories.

Project 3:

Project: Bedrock (AWS Client)
Role: Data Scientist & Team Manager (Different PODs)
Impact & Achievement Focus
Large Language Model Training Data: Managed and guided a 10-person data science multiple teams in the critical review and refinement of over 50,000 high-quality prompts for training Amazon Web Services' foundational Large Language Models (LLMs).
Quality and Compliance: Established and enforced a plagiarism-free and grammatically rigorous content policy, accelerating the LLM fine-tuning schedule by an estimated 2 months.
Performance and Metrics: Developed and maintained the Python-based internal Leaderboard and Key Performance

Indicators (KPIs) system, ensuring the prompt-ranking team consistently exceeded daily quality targets.

Project Title: Experience monitoring

Project: Experience Monitoring (Dell Client)

Technologies: Python, Pandas, Machine Learning, Flask API

Impact & Achievement Focus

Predictive Modelling: Engineered and deployed a regression model to predict manufacturing cycle duration for individual orders across 7 distinct stages (Build, Network, Shuttle, etc.).

Quantified Performance: The model achieved a 15% reduction in Mean Absolute Error (MAE) compared to the previous baseline, directly contributing to a 5% reduction in the overall product manufacturing cycle time.

Data Engineering: Led the complex feature engineering and data preparation for one year of production data consolidated from disparate sources (Teradata, Oracle), supporting all subsequent modelling efforts.

Sr Technical Lead and Managing team, Data science (SureIT Solutions August 2020- September 2021)

Project: Falcon Project (Johnson & Johnson Client)

Technologies: Python, Pandas, Machine Learning

Anomaly Detection: Built and deployed a supervised machine learning model that achieved 97% accuracy in predicting job failures and anomalies across various servers.

Time Prediction Modelling: Developed two separate machine learning models to accurately predict the completion time for AG ETL and IRT ETL jobs, improving resource scheduling reliability by 12%.

Project Title: Central Clearing House (CCH)

Project Description:

The project related to transportation department. Identifying the vehicle number plates which are travelling on different agencies and hubs in US. whenever any vehicle travel on roads of agencies identifying corresponding details based on number plate. Scheduling different files by using airflow. Transforming different formats of files from one agency to another agency.

Project: Central Clearing House (CCH)

Technologies: Python, NLP, OpenCV, Flask API

Image Processing & NLP: Developed an automated pipeline using OpenCV and Natural Language Processing (NLP) to identify and process vehicle number plate details from different US agencies, achieving 95% accuracy in plate recognition.

System Architecture: Led the architectural design for the project, integrating Airflow for file scheduling and deploying the core models via a high-availability Flask API.

Project Title: Nightingale (Pyramid IT Solutions Jan 2020 – May 2020)

Project: Virtual Nursing Assessment

Technologies: Python, ML/DL, NLP, Flask API

Critical Care Prediction: Developed a predictive model using the Rothman Index analysis and 26 physiological parameters to identify patients at risk of mortality within 24 hours, achieving 94% accuracy in early warning alerts.

Diabetes & Heart Disease Modelling: Applied various machine learning algorithms to achieve 90% accuracy in classifying Type 1/Type 2 diabetes and performing heart disease risk analysis.

Usability & Sentiment Analysis: Implemented Natural Language Processing (NLP) for patient sentiment analysis on

feedback data, providing insights that drove a 15% improvement in the Virtual Nursing interface design.

Technical Leadership: Guided 3 junior data scientists on best practices in model selection and high-volume patient data analysis.

Unisys, Hyderabad, Telangana

Senior Software Engineer, Data science

(Dec 2017-Nov 2019)

Project: Civil Registry System (Philippines Government Client)

Technologies: Python, ML/DL, NLP, Chatbot, Tableau

Imputation Modelling: Developed and deployed an imputation model that predicted missing age values in the vital events data (birth, marriage, death) with 96% accuracy, streamlining the government certificate issuance process.

Predictive Analysis: Used machine learning to predict customer behaviour and demand for various certificates across different cities, enabling the client to optimize resource allocation and staff recruitment, resulting in a 10% reduction in document processing time.

AI/NLP Implementation: Designed and developed a Deep Learning-based Chatbot that successfully handled 80% of routine customer inquiries, significantly reducing call centre load.

Data Analytics: Led the design of Tableau dashboards for executive reporting, visualizing trends in 100,000+ annual certificate issuances by quarter, year, and city.

Honeywell, Bangalore, Karnataka (Feb 2017 – Dec 2017)

Project: Aerospace Estimation Tool (AET)

Technologies: Python, Machine Learning, NLTK

Budget Forecasting: Developed and implemented a machine learning regression model for budget plan estimation, reducing forecasting variance by 8% compared to traditional estimation methods.

Text Analysis: Utilized NLTK and machine learning for sentiment analysis on 500+ employee performance reviews, categorizing performance risk and providing actionable insights to management.

Client: Contractor at ITC InfoTech, Bangalore, Karnataka (Sept 2016 – Feb 2017)

Project Title: Escalator Calculation Tool

Project: Escalator Calculation Tool (KONE Client)

Technologies: Python, Machine Learning (Supervised/Unsupervised)

Energy Optimization: Developed a supervised machine learning model to accurately calculate and optimize Escalator Energy Consumption, leading to component selection efficiency gains of 7%.

Requirements Prediction: Used machine learning to predict Hand Rail Requirements and component calculations based on client specifications, accelerating the proposal generation cycle by 1 week.

Software Engineer

Project Title: Healthipass (Magus IT Solutions July 2012- July 2016)

Project: Healthipass Product

Technologies: Timeseries Analysis, Data Analysis

Appointment Scheduling: Implemented Timeseries Analysis to optimize appointment scheduling logic, reducing patient wait times by an average of 15 minutes.

Insurance Behaviour: Analysed health insurance claim patterns to provide patients with upfront coverage benefit details,

which helped clinics minimize losses from unpaid balances.

Teaching Experience

Assistant Professor, St. John's College of Engg & Tech (June 2009 – Nov 2010)

- Taught core Computer Science subjects including Machine Learning, Python, and Data Structures, C programming & data structures, Compiler Design, Theory of computation, Formal languages Automata Theory, Automat Theory and Compiler Design.
- Guided 15+ student mini- and major-projects.

Corporate Training & Scientific Leadership

- **Corporate Training:** Trained **1,000+ Deloitte professionals** on Python and Machine Learning technologies, with courses achieving an average satisfaction score of **4.8/5.0** and a 90-day skill adoption rate of **70%**.
- **Associate Editor:** Acted as Associate Editor for IEEE Transactions on Artificial Intelligence 2023-2024, demonstrating high-level scientific and editorial judgment.
- **Reviewer:** Serving as a reviewer for 20+ high-impact SCI and Scopus journals (e.g., Neurocomputing, Progress in Artificial Intelligence (PRAI), Intelligent Service Robotics (JIST), Engineering Applications of Artificial Intelligence, Pattern Recognition, Discover Computing, Scientific Reports, Discover Applied Sciences, Discover Internet of Things, Evolving Systems, Heliyon, International Journal of Data Science and Analytics, Computers and Electronics in Agriculture, Decision Analytics Journal, Signal, Image and Video Processing, Journal of Transformative Technologies and Sustainable Development, Multimedia Tools and Applications, IEEE Access , SN Computer Science , Discover Environment, Plant-Environment Interactions, Review for Chemometrics and Intelligent Laboratory Systems).

Invited Talks and Lectures:

This section highlights my contributions to academic and professional knowledge transfer, emphasizing expertise in contemporary AI topics.

Date/Duration	Host Institution/Audience	Topic	Focus & Methodology	Outcome/Impact
Jan 2024	Nitte Meenakshi Institute of Technology (3rd Year CSE)	Workshop: AI and Machine Learning	Comprehensive 1-day workshop covering supervised/unsupervised learning, model evaluation metrics, and the transition from classical ML to Deep Learning. Emphasized practical implementation using Python libraries.	Equipped 80+ participants with the ability to prototype basic ML models and understand the trade-offs between algorithm complexity and interpretability.
Mar 2024	Sreenidhi Institute of Science and Technology (3rd Year CSE)	Guest Lecture: Core Machine Learning	Focused session on advanced ML concepts, including ensemble methods, regularization techniques, and the importance of Explainable AI (XAI) principles	Provided students with insight into ML maturity models and how industry-standard XAI techniques drive model trust and adoption in critical domains.
Jun 2024	Sreenidhi Institute of Science and Technology (2nd Year CSE)	Guest Lecture: Python Programming for Data Science	A practical introduction to Python for data handling, focusing on Pandas for data manipulation and Matplotlib/Seaborn for exploratory data analysis, setting the foundation for ML coursework.	Ensured a strong prerequisite skill foundation for students transitioning into advanced data science and AI curricula, improving subsequent course performance.

Feb 2025	St. Johns College of Engineering & Technology (3-day Workshop)	Workshop: Python Programming & Data Structures	Intensive 3-day workshop bridging academic CS theory with practical Python implementation for high-volume data structures, a crucial foundation for efficient AI model development.	Elevated the programming fluency of 100+ students, directly supporting their ability to execute B. Tech major projects requiring substantial algorithmic complexity.
Feb-26	CBIT, Hyderabad	Workshop: Opening the Blackbox	A practical introduction to XAI with various use cases	Elevated the programming fluency with XAI
April - 26	RGM, Nandyal	Python for AI Era	A practical usecases how python can be used in AI programming	Elevated the programming fluency with python
Mar-26	NBKR, Nellore	Workshop on GenAI	A practical introduction to GenAI with various use cases	Elevated the programming fluency with GenAI
June - 26	CBIT, Hyderabad	International FDP	The Agentic Classroom: Redefining Pedagogy with Generative AI and Intelligent Agents	Elevated the programming fluency with GenAI and agentic AI